Rui Zhu

Applied Scientist II, Amazon
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Google Scholar

Research Interests

DNA large language models; Security, safety and privacy of large language models.

Education

2019 – 2024 **Ph.D., Indiana University Bloomington**, Department of Computer Science

2017 – 2019 **M.S., Indiana University Bloomington**, Department of Statistics

2011 – 2015 **B.S., University of Utah**, Department of Mathematics

Work Experience

2026 Mar – Now **Applied Scientist II** in Amazon;
Medical Foundation Model; Built and trained a medical foundation model from scratch with multiple modules integrating heterogeneous data sources, including EHR records, claims data, drug SMILES, and patient genomic data.)

Affiliated Lecturer in Yale School of Medicine;

2024 Jun – 2026 Feb **Post-doctoral Researcher** in Yale University, School of Medicine, *Advisor: Dr. Lucila Ohno-Machado*;
DNA Large Language Model; Designed downstream applications based on a DNA large language model (Evo2): performed continued pretraining on human pan-genome data to "humanize" the model for human genetic variation, and supervised fine-tuning on disease-labeled human genomics datasets for clinical prediction tasks.
LLM Model Compression; Implemented model compression for the DNA LLM (Evo2) to enable more efficient inference on much longer genomic sequences.
Tabular Large Foundation Model; Built applications on top of the tabular foundation model TabPFN by fine-tuning it on clinical datasets, achieving performance that outperforms standard gradient-boosting–based methods.

2023 May-August **Software Engineer** in The Trade Desk
Adversarial Model; Developed a deep learning–based adversarial model to predict competitors (such as Google Ads) bidding for on-page impressions.

2018 May-August **Data Scientist** Union Pay Lab
Community detection; To build this community detection model. One patent was applied in this project(Patent number: CN09493086A, Patent page: http://www.soopat.com/Patent/201811261870).

2016 – 2017 **Data Scientist** in Lenovo;
Text Classification; Text classification model for online sentiment analysis of customer comments in e-commerce.
Fake Comments Detection in E-commerce; Identify whether a comment from e-commerce is from a real user or a machine.
Retailer Fraud Detection; To detect Insurance and warranty frauds in Retailers of Lenovo.

Teaching Experience

Lecturer — **I599: Data Science on-ramp** – Fall 2019 - Fall 2020 (3 semesters)
I529: Machine Learning in Bioinformatics – Fall 2023 - Spring 2024

Achievements

Apr 2023 — **Indiana University Bloomignton**: Best Research Assistant of the year 2022 in computer science department

Patent

community detection model

2018.11 — Patent number: CN09493086A, Patent page: http://www.soopat.com/Patent/201811261870

Projects

TrojAI project by NIST and IARPA

2020.5 – 2024.5 — In team Perspectra-IUB, Main contributor. Main weekly meeting reporter. Got 2 out of 16 teams in round 3 competition and 3 out of 16 teams in round 10 competition.

Publications

** indicates co-first authors.*

Highlight Manuscripts

1. Wang*, Z., **Zhu*, Rui**, Zhou, D., Zhang, Z., Wang, X., & Tang, H. (2025). Sharpness-aware initialization: Improving differentially private machine learning from first principles. *34th USENIX Security Symposium (USENIX Security 2025)*, 3103–3122.
2. Xiong, C., Wang, Z., **Zhu, Rui**, Ho, T.-Y., Chen, P.-Y., Xiong, J., Tang, H., & Ohno-Machado, L. (2025). Hey, that's my data! label-only dataset inference in large language models. *arXiv preprint arXiv:2506.06057 (ACL 2026 Accepted)*.
3. Chen*, X., Tang*, S. Z. R., Yan, S., Jin, L., Wang, Z., Su, L., Zhang, Z., Wang, X., & Tang, H. (2024). The janus interface: How fine-tuning in large language models amplifies the privacy risks. *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 1285–1299.
4. Wang*, Z., **Zhu*, Rui**, Zhou, D., Zhang, Z., Mitchell, J., Tang, H., & Wang, X. (2024). Dpadapter: Improving differentially private deep learning through noise tolerance pre-training. *USENIX Security*.

Trustworthy AI and Data

1. Fan*, Y. Z. R., Wang, Z., Wang, C., Tang, H., Dong, Y., Cho, H., & Ohno-Machado, L. (2025). Byzsfl: Achieving byzantine-robust secure federated learning with zero-knowledge proofs. *arXiv preprint arXiv:2501.06953*.
2. Li, A., Qian, L., Du, M., Yin, Y., Hu, Y., **Zhu, Rui**, Sun, Z., Fu, Y., Stutz, E., Ai, X., Xie, Q., et al. (2025). Memorization in large language models in medicine: Prevalence, characteristics, and implications. *arXiv preprint arXiv:2509.08604 (Nature Communication Accepted)*.
3. Li, X., Song, K., **Zhu, Rui**, Chen, P.-Y., & Tang, H. (2025). Adversarial attack-defense co-evolution for llm safety alignment via tree-group dual-aware search and optimization. *arXiv preprint arXiv:2511.19218*.

4. **Zhu, Rui**, Mortensen, G., Fan, Y., & Tang, H. (2025). Illuminating the unseen: A large-scale exploration of bias in icu discharge summaries via language models. *IEEE BHI*.

5. Wang*, Z., **Zhu*, Rui**, Zhang, Z., Tang, H., & Wang, X. (2025). Rigging the foundation: Manipulating pre-training for advanced membership inference attacks. *IEEE S&P*.

6. Wang*, Z., **Zhu*, Rui**, Zhou, D., Zhang, Z., Wang, X., & Tang, H. (2025). Sharpness-aware initialization: Improving differentially private machine learning from first principles. *34th USENIX Security Symposium (USENIX Security 2025)*, 3103–3122.

7. Xiong, C., Wang, Z., **Zhu, Rui**, Ho, T.-Y., Chen, P.-Y., Xiong, J., Tang, H., & Ohno-Machado, L. (2025). Hey, that's my data! label-only dataset inference in large language models. *arXiv preprint arXiv:2506.06057 (ACL 2026 Accepted)*.

8. Chen*, X., Tang*, S. Z. R., Yan, S., Jin, L., Wang, Z., Su, L., Zhang, Z., Wang, X., & Tang, H. (2024). The janus interface: How fine-tuning in large language models amplifies the privacy risks. *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 1285–1299.

9. Tang, D., **Zhu, Rui**, Chen, Y., Tang, H., & Wang, X. (2024). Understanding impacts of task similarity on backdoor attack and detection. *arXiv preprint arXiv:2210.06509*.

10. **Zhu, Rui**, Tang, D., Tang, S., Wang, Z., Tao, G., Ma, S., Wang, X., & Tang, H. (2024). Gradient shaping: Enhancing backdoor attack against reverse engineering. *NDSS*.

11. Wang*, Z., **Zhu*, Rui**, Zhou, D., Zhang, Z., Mitchell, J., Tang, H., & Wang, X. (2024). Dpadapter: Improving differentially private deep learning through noise tolerance pre-training. *USENIX Security*.

12. Li*, Z., **Zhu*, Rui**, Wang, Z., Tang, H., & Chai, H. (2023). Fairfix: Improving fairness of deep neural networks through selective amnesia and recovery. *SmartCom (Best Paper Award)*.

13. **Zhu*, Rui**, Tang*, D., Tang, S., Wang, X., & Tang, H. (2023). Selective amnesia: On efficient, high-fidelity and blind suppression of backdoor effects in trojaned machine learning models. *IEEE S&P*.

14. **Zhu, Rui**, Jiang, C., Wang, X., Wang, S., Zheng, H., & Tang, H. (2020). Privacy-preserving construction of generalized linear mixed model for biomedical computation. *Bioinformatics, ISMB2020, 36*(Suppl 1), i128.

AI for Science

1. **Zhu, R.**, Zhou, X., Liang, I., Stephen, S., Xu, K., & Ohno-Machado, L. (2026a). Foundation model–guided synthetic ehr release: Performance enhancement with privacy preservation. *AMIA Amplify Informatics Summit*.

2. **Zhu, R.**, Zhou, X., Liang, I., Stephen, S., Xu, K., & Ohno-Machado, L. (2026b). Large models for small tables: Adapting tabular foundation models to ehr data. *AMIA Amplify Informatics Summit*.

3. Ge, C., Xiong, J., **Zhu, Rui**, Hong, Z., & He, Y. (2025a). The global burden of high bmi among adolescents between 1990 and 2021. *Communications Medicine*, *5*(1), 125.

4. Ge, C., Xiong, J., **Zhu, Rui**, Hong, Z., & He, Y. (2025b). Global burden of high sugar-sweetened beverage consumption among young adults. *Diabetology & Metabolic Syndrome*, *17*(1), 259.

5. Ma*, X., **Zhu, Rui***, Wang, Z., Xiong, J., Chen, Q., Tang, H., Camp, L. J., & Ohno-Machado, L. (2025). Enhancing patient-centric communication: Leveraging llms to simulate patient perspectives. *arXiv preprint arXiv:2501.06964*.

6. Mortensen*, G. Z. R. (2025). Adetectolocum: Harnessing locally deployable llms for early alzheimer's disease detection through voice analysis. *AMIA Informatics Summit 2025*.

7. **Zhu*, Rui**, Zhou*, X., Aquino, M. M. d., Engchuan, W., Zhou, H., Cheng, H., Initiative, A. D. N., Hardy, J., Tang, H., Scherer, S. W., et al. (2025). Advancing human population genomics with dna foundation models. *medRxiv (Nature Communication under review)*, 2025–09.

8. **Zhu, Rui**, Zhou, X., Tang, H., Scherer, S. W., & Ohno-Machado, L. (2025). Near-lossless model compression enables longer context inference in dna large language models. *arXiv preprint arXiv:2511.14694 (RECOMB 2026 under review).*

9. Zhang, K., **Zhu, Rui**, Ma, S., Xiong, J., Kim, Y., Murai, F., & Liu, X. (2025). Kedrec-lm: A knowledge-distilled explainable drug recommendation large language model. *NIPIR Workshop*.

Others

1. Li, X., Jiao, W., Song, K., **Zhu, Rui**, & Tang, H. (2025). Knowledge-aware co-reasoning for multidisciplinary collaboration. *EMNLP*.

2. Li, X., Song, K., **Zhu, Rui**, Tang, H., et al. (2025). Knowledge-aware co-reasoning for multidisciplinary collaboration. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 13615–13631.

3. Li, X., Liu, K., **Zhu, Rui**, & Liu, X. (2023). Hierarchical multi-task learning for enterprise risk detection from financial documents. *IEEE BigData*.

4. Li, X., Qin, Y., **Zhu, Rui**, Lin, T., Fan, Y., Kang, Y., Song, K., Zhao, F., Sun, C., Tang, H., & Liu, X. (2023). Stinmatch: Semi-supervised semantic-topological iteration network for financial risk detection via news label diffusion. *EMNLP*.

5. **Zhu, Rui**, Lin, B., & Tang, H. (2020). Bounding the number of linear regions in local area for neural networks with relu activations. *arXiv preprint arXiv:2007.06803*.

Media Coverage

2023

- **The New York Times**: Interviewed by NYT journalist Jeremy White and featured in the front-page story "How strangers got my email address from ChatGPT's model" `https://www.nytimes.com/interactive/2023/12/22/technology/openai-chatgpt-privacy-exploit.html`
- **Business Today**: "Email used to log in for ChatGPT at potential risk, research team unveils OpenAI's GPT-3.5 Turbo threat" `https://www.businesstoday.in/technology/news/story/email-used-to-log-in-for-chatgpt-at-potential-risk-research-team-unveils-openais-gpt-35-turbo-threat-410774-2023-12-26`

Service To Professional Community

- **Served as Subject Matter Expert**: Patient-Centered Outcomes Research Institute (PCORI)
- **Served as Associate Editor**: IEEE Engineering in Medicine and Biology Society (IEEE EMBC)
- **Served as Chair**: NDSS Worshop on AI System with Confidential Computing (AISCC 2024)
- **Served as Organizer**: integrating Data for Analysis, Anonymization and Sharing (iDASH 2024)
- **Served as PC**: International Conference on Database Systems for Advanced Applications (DASFAA 2024)
- **Served as PC**: ACM Conference on Data and Application Security and Privacy (CODASPY)
- **Served as Reviewer**: Bioinformatics Advances
- **Served as Reviewer**: IEEE Transactions on Information Forensics & Security
- **Served as PC**: ACM Conference on Data and Application Security and Privacy (CODASPY)
- **Served as Reviewer**:IEEE Journal of Biomedical and Health Informatics
- **Served as Artifact PC member**: Annual Computer Security Applications Conference (ACSAC)
- **Served as Organizer and PC**:Backdoor Attacks aNd DefenSes in Machine Learning (BANDS) Workshop at ICLR 2023

Service To Professional Community (continued)

- **Served as Reviewer**:Intelligent Systems for Molecular Biology
- **Served as PC member**: Bioinformatics journal
- **Served as Artifact PC member**: Annual Computer Security Applications Conference (ACSAC)
- **Served as PC member and session Chair**: European Symposium on Research in Computer Security
- **Served as PC member and session Chair**: European Symposium on Research in Computer Security